

EDUCATION

University of Prince Edward Island

B.Sc. in Computer science, Minor in Economics

- Data Structures and Algorithms, Software Engineering, Operating Systems, System Design, Machine Learning.

PROJECTS

Monopoly Strategy Simulation | Java

- Designed and implemented a simulation engine modelling financial transaction, property management, and strategic decision-making.
- Developed four AI-driven strategies and executed 60+ simulations to evaluate performance across metrics (win rate, balance, asset distribution).
- Applied concepts similar to multithreaded simulations and portfolio back-testing.

MapleNest | Node.js, Express, MongoDB, React, Tailwind CSS

- Designed and developed a full-stack web application that aggregates long-term rentals and short-term sublets across Canada, targeted at students and young professionals.
- Built RESTful APIs using Node.js and Express to collect, normalize, and serve housing data with filtering by location, price range, and rental type.
- Implemented a MongoDB data model to efficiently store listings and support scalable search and filtering operations.
- Developed a responsive, mobile-first frontend in React and Tailwind CSS to enable intuitive browsing and fast user interactions.
- Structured the application to support future automation such as scheduled data updates, saved searches, and alerting features.

UNBORED | Next.js, Tailwind CSS, React, REST API

- Developed a responsive real-time messaging platform with thread-based conversations and individual message histories.
- Designed a **mobile-first UI** with Tailwind CSS, ensuring accessibility and usability across devices.
- Structured application state management to handle concurrent message updates without data conflicts or UI lag.
- Resolved real-time messaging sync issues by tracing API call failures, analyzing logs, and implementing fixes to prevent race conditions.

EXPERIENCE

Software Engineer Intern

British Columbia Electronic Library Network

Jan 2025 – Apr 2025

- Investigated technical problems reported by clients, collaborating directly with software engineers, product owners, and data librarians to identify root causes and implement fixes.
- Created internal documentation outlining system workflows, troubleshooting steps, and integration processes to support both technical and non-technical users.
- Managed multiple concurrent tasks in an Agile environment, contributing to unit testing, debugging, and iterative improvements to data-driven applications.
- Integrated the TK Labels API, applying semantic mapping techniques and documenting lessons learned for future API-onboarding projects.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, Dart, C++, TypeScript, PHP.
- **Frameworks & Tools:** React, Node.js, JUnit, Spring, Git, Linux, VS Code
- **Databases & Infrastructure:** SQL, Redis, MongoDB, AWS, Docker, REST APIs